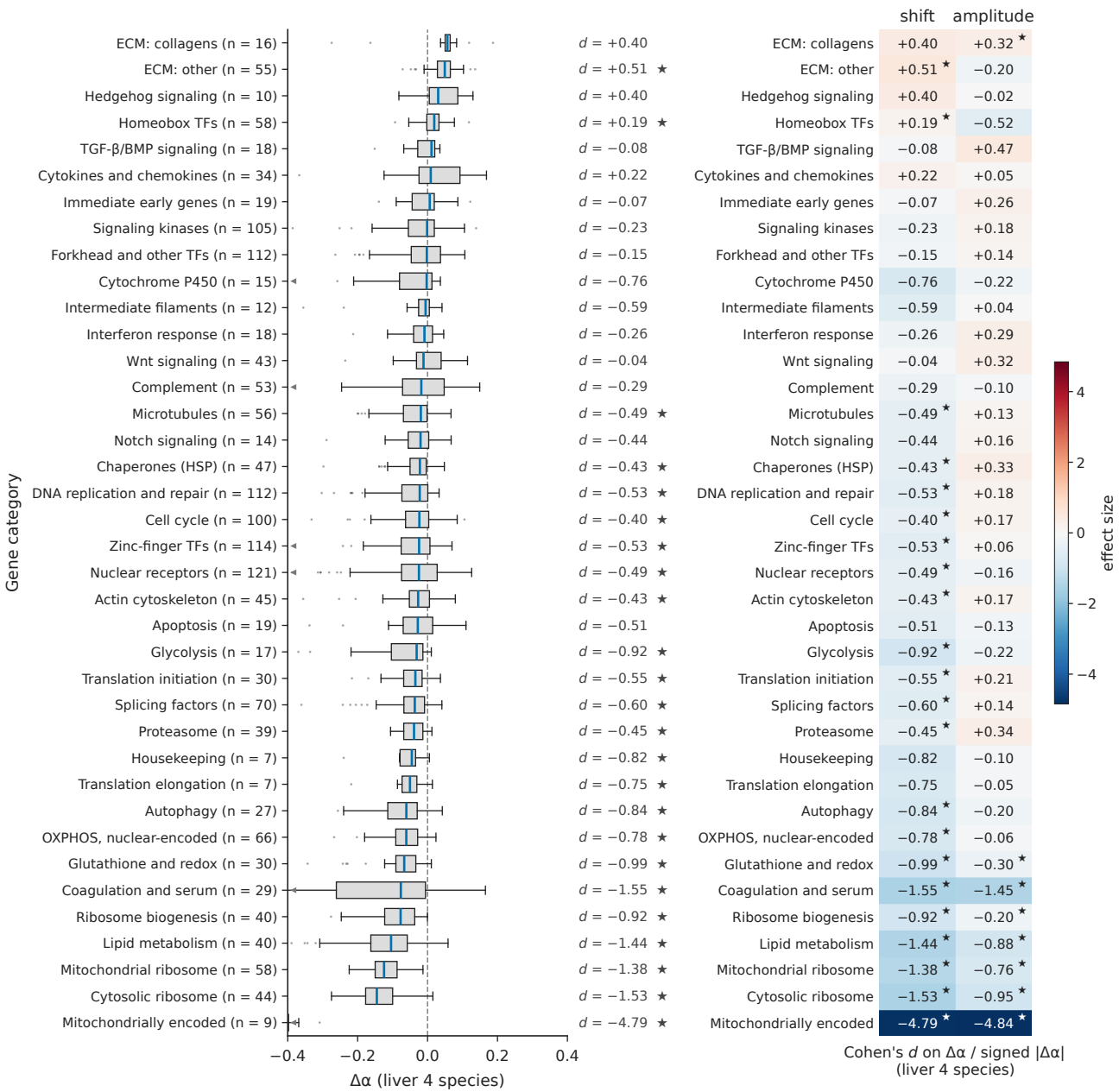

Gene category
ECM: collagens (n = 16)
ECM: other (n = 55)
Hedgehog signaling (n = 10)
Homeobox TFs (n = 58)
TGF-β/BMP signaling (n = 18)
Cytokines and chemokines (n = 34)
Immediate early genes (n = 19)
Signaling kinases (n = 105)
Forkhead and other TFs (n = 112)
Cytochrome P450 (n = 15)
Intermediate filaments (n = 12)
Interferon response (n = 18)
Wnt signaling (n = 43)
Complement (n = 53)
Microtubules (n = 56)
Notch signaling (n = 14)
Chaperones (HSP) (n = 47)
DNA replication and repair (n = 112)
Cell cycle (n = 100)
Zinc-finger TFs (n = 114)
Nuclear receptors (n = 121)
Actin cytoskeleton (n = 45)
Apoptosis (n = 19)
Glycolysis (n = 17)
Translation initiation (n = 30)
Splicing factors (n = 70)
Proteasome (n = 39)
Housekeeping (n = 7)
Translation elongation (n = 7)
Autophagy (n = 27)
OXPHOS, nuclear-encoded (n = 66)
Glutathione and redox (n = 30)
Coagulation and serum (n = 29)
Ribosome biogenesis (n = 40)
Lipid metabolism (n = 40)
Mitochondrial ribosome (n = 58)
Cytosolic ribosome (n = 44)
Mitochondrially encoded (n = 9)
d = +0.40
d = +0.51 ★
d = +0.40
d = +0.19 ★
d = −0.08
d = +0.22
d = −0.07
d = −0.23
d = −0.15
d = −0.76
d = −0.59
d = −0.26
d = −0.04
d = −0.29
d = −0.49 ★
d = −0.44
d = −0.43 ★
d = −0.53 ★
d = −0.40 ★
d = −0.53 ★
d = −0.49 ★
d = −0.43 ★
d = −0.51
d = −0.92 ★
d = −0.55 ★
d = −0.60 ★
d = −0.45 ★
d = −0.82 ★
d = −0.75 ★
d = −0.84 ★
d = −0.78 ★
d = −0.99 ★
d = −1.55 ★
d = −0.92 ★
d = −1.44 ★
d = −1.38 ★
d = −1.53 ★
d = −4.79 ★
−0.4
−0.2
0.0
0.2
0.4
Δα (liver 4 species)
(a)
shift
amplitude
ECM: collagens +0.40 +0.32 ★
ECM: other +0.51 ★ −0.20
Hedgehog signaling +0.40 −0.02
Homeobox TFs +0.19 ★ −0.52
TGF-β/BMP signaling −0.08 +0.47
Cytokines and chemokines +0.22 +0.05
Immediate early genes −0.07 +0.26
Signaling kinases −0.23 +0.18
Forkhead and other TFs −0.15 +0.14
Cytochrome P450 −0.76 −0.22
Intermediate filaments −0.59 +0.04
Interferon response −0.26 +0.29
Wnt signaling −0.04 +0.32
Complement −0.29 −0.10
Microtubules −0.49 ★ +0.13
Notch signaling −0.44 +0.16
Chaperones (HSP) −0.43 ★ +0.33
DNA replication and repair −0.53 ★ +0.18
Cell cycle −0.40 ★ +0.17
Zinc-finger TFs −0.53 ★ +0.06
Nuclear receptors −0.49 ★ −0.16
Actin cytoskeleton −0.43 ★ +0.17
Apoptosis −0.51 −0.13
Glycolysis −0.92 ★ −0.22
Translation initiation −0.55 ★ +0.21
Splicing factors −0.60 ★ +0.14
Proteasome −0.45 ★ +0.34
Housekeeping −0.82 −0.10
Translation elongation −0.75 −0.05
Autophagy −0.84 ★ −0.20
OXPHOS, nuclear-encoded −0.78 ★ −0.06
Glutathione and redox −0.99 ★ −0.30 ★
Coagulation and serum −1.55 ★ −1.45 ★
Ribosome biogenesis −0.92 ★ −0.20 ★
Lipid metabolism −1.44 ★ −0.88 ★
Mitochondrial ribosome −1.38 ★ −0.76 ★
Cytosolic ribosome −1.53 ★ −0.95 ★
Mitochondrially encoded −4.79 ★ −4.84 ★
effect size
4
2
0
−2
−4
Cohen's d on Δα / signed |Δα|
(liver 4 species)
(b)